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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

Vehicle Rental Booking and Payment Management System

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DECLARATION

I hereby declare that the project entitled "**Vehicle Rental Booking and Payment Management System**" submitted in partial fulfillment of the requirements for the course [**Course Name]** is a **record of my original work** carried out under the guidance of **[Trainer Name]**, Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

KAVYA S

CERTIFICATE

This is to certify that the project entitled "**Vehicle Rental Booking and Payment Management System"** is a **bona fide work** carried out by **[KAVYA S] [1-ABatch]**, [**BSc Information Technology]**, **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **[Course Name]**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place:

Date:

Submitted for the Viva-Voce Examination held on _____

Guide:
(Technical Trainer, KGCAS)

Head of the Department
(Technical Head, KGCAS)

TABLE OF CONTENTS

S. No.	Topics	Page. No.
1	**Introduction**	
1.1	Objective	
1.2	Overview	
1.3	Features	
2	**Program Analysis**	
2.1	Problem Definition	
2.2	Project Overview	
2.3	Scopes and Limitations	
3	**System Design**	
3.1	Modules	
4	**Methodology**	
4.1	Problem Analysis	
4.2	System Design	
4.3	Program Development	
4.4	Testing and Validation	
4.5	Result and Documentation	
4.6	Algorithm	
4.7	Flowchart	
5	**Source Code**	
6	**Input and Output Screens**	
7	**Discussion and Conclusion**	
7.1	Discussion	
7.2	Conclusion	
8	**References**	

ABSTRACT

The Vehicle Rental Booking and Payment Management System is a comprehensive Java-based application designed to modernize, automate, and streamline the core operations of vehicle rental agencies. In the current fast-paced transportation industry, many small to medium-sized rental businesses continue to rely on manual, paper-based methods for managing their fleet and processing customer transactions. These traditional practices are inherently inefficient, often leading to significant human errors in calculating rental durations, misplaced records, and delays in generating customer invoices. The primary objective of this project is to replace these outdated manual logs with a precise, digital framework that enhances operational accuracy and improves overall customer satisfaction.

This system provides a centralized digital platform where administrative staff can manage a diverse inventory of vehicles, including economy cars, luxury sedans, SUVs, and two-wheelers. The application allows users to view a real-time list of available vehicles, select a specific model based on customer preference, and input the exact rental duration in days. Once the selection is finalized, the system's core logic automatically calculates the subtotal based on predefined daily rates and applies mandatory taxes, such as GST, to generate a transparent and professional final bill. This automation ensures that mathematical discrepancies are eliminated, providing both the business and the customer with an accurate reflection of the costs involved.

The application is developed using Core Java programming concepts, utilizing an Object-Oriented Programming (OOP) approach to ensure code modularity and scalability. Key technical components implemented in this system include:

- Classes and Objects: Used to model the vehicle fleet and individual rental transactions.
- Methods: Developed to handle specific tasks such as menu navigation, cost calculation, and invoice formatting.
- Conditional Statements and Loops: Utilized to manage user choices and process multiple vehicle selections within a single booking session.
- Exception Handling: Integrated to manage invalid inputs gracefully, ensuring the system remains stable during user interaction.

Ultimately, the Vehicle Rental Booking and Payment Management System demonstrates the practical application of Java programming in solving real-world business challenges. By reducing the time required for bill generation and ensuring the integrity of financial records, the system serves as an essential tool for modern rental agencies. While the current version is optimized as a robust console-based application, the architecture is designed to support future

enhancements, including database integration for long-term data persistence and the development of a Graphical User Interface (GUI) for an even more intuitive user experience.

INTRODUCTION

The Vehicle Rental Booking and Payment Management System is a Java-based web application designed to streamline vehicle rentals, featuring online booking, fleet management, and secure payment processing. It enables users to browse, book, and pay for vehicles, while providing administrators with tools to manage inventory, customers, and transactions, ultimately reducing operational labor.

1.1 Objective

The primary objective of this system is to automate the traditional manual vehicle rental process. It aims to provide a centralized platform where customers can browse available vehicles, make real-time bookings, and execute secure digital payments, while allowing administrators to track fleet health and revenue.

1.2 Overview

This system acts as a bridge between car rental agencies and users. It utilizes a web/mobile interface to handle high-concurrency booking requests, ensuring that no vehicle is double-booked and that all transactions are logged for financial auditing.

1.3 Features

Real-time Availability: Instant updates on vehicle status.

Dynamic Pricing: Rates based on vehicle type and duration.

Secure Payment Gateway: Integration with credit cards and digital wallets.

User Profiles: History of rentals and saved payment methods.

Admin Dashboard: Fleet management and analytics.

2. Program Analysis

2.1 Problem Definition

Traditional rental systems often suffer from paperwork delays, lack of transparency in pricing, and high risks of overbooking. Physical payment handling also increases the risk of accounting errors.

2.2 Project Overview

The project is a multi-tier application. It consists of a frontend for user interaction, a backend for business logic (handling the booking engine), and a database to store user, vehicle, and transaction data.

2.3 Scopes and Limitations

Scope: Includes booking, payment, and basic fleet tracking.

Limitations: Does not include hardware-based GPS tracking in this version; requires internet connectivity for real-time validation.

3. System Design

3.1 Modules

User Management: Registration, Login, and Profile Management.

Vehicle Management: Adding, editing, or removing cars from the fleet.

Booking Engine: The core logic that calculates dates, availability, and total costs.

Payment Module: Handling API calls to payment processors.

Review/Rating System: Post-rental feedback loop.

4. Methodology

4.1 Problem Analysis

We utilize the SDLC (Software Development Life Cycle). The first phase involves gathering requirements from stakeholders (fleet owners and frequent travelers) to identify pain points like hidden fees and slow pickup times.

4.2 System Design

The architecture follows the MVC (Model-View-Controller) pattern. This separates the data (Model) from the user interface (View) and the logic (Controller), making the system easier to scale and maintain.

4.3 Program Development

Development is carried out in an Agile environment. Sprints focus on specific features (e.g., "Sprint 1: Database Setup," "Sprint 2: Booking Logic"). Popular stacks include MERN (MongoDB, Express, React, Node) or LAMP.

4.4 Testing and Validation

Unit Testing: Testing individual functions like price calculations.

Integration Testing: Ensuring the booking module correctly updates the database.

User Acceptance Testing (UAT): Real users testing the flow from selection to payment.

4.5 Result and Documentation

The final product is a functional system that reduces booking time by an estimated 70% compared to manual processes. Documentation includes API endpoints and user manuals.

4.6 Algorithm

A critical algorithm for this system is the Availability Checker:

Input: Requested Date Range (D

start

,D

end

) and Vehicle ID.

Query database for existing bookings for that ID.

If any existing booking overlaps (D

start

<B

end

and D

end

>B

start

), return "Unavailable."

Else, return "Available" and calculate price: Total=(D

end

−D

start

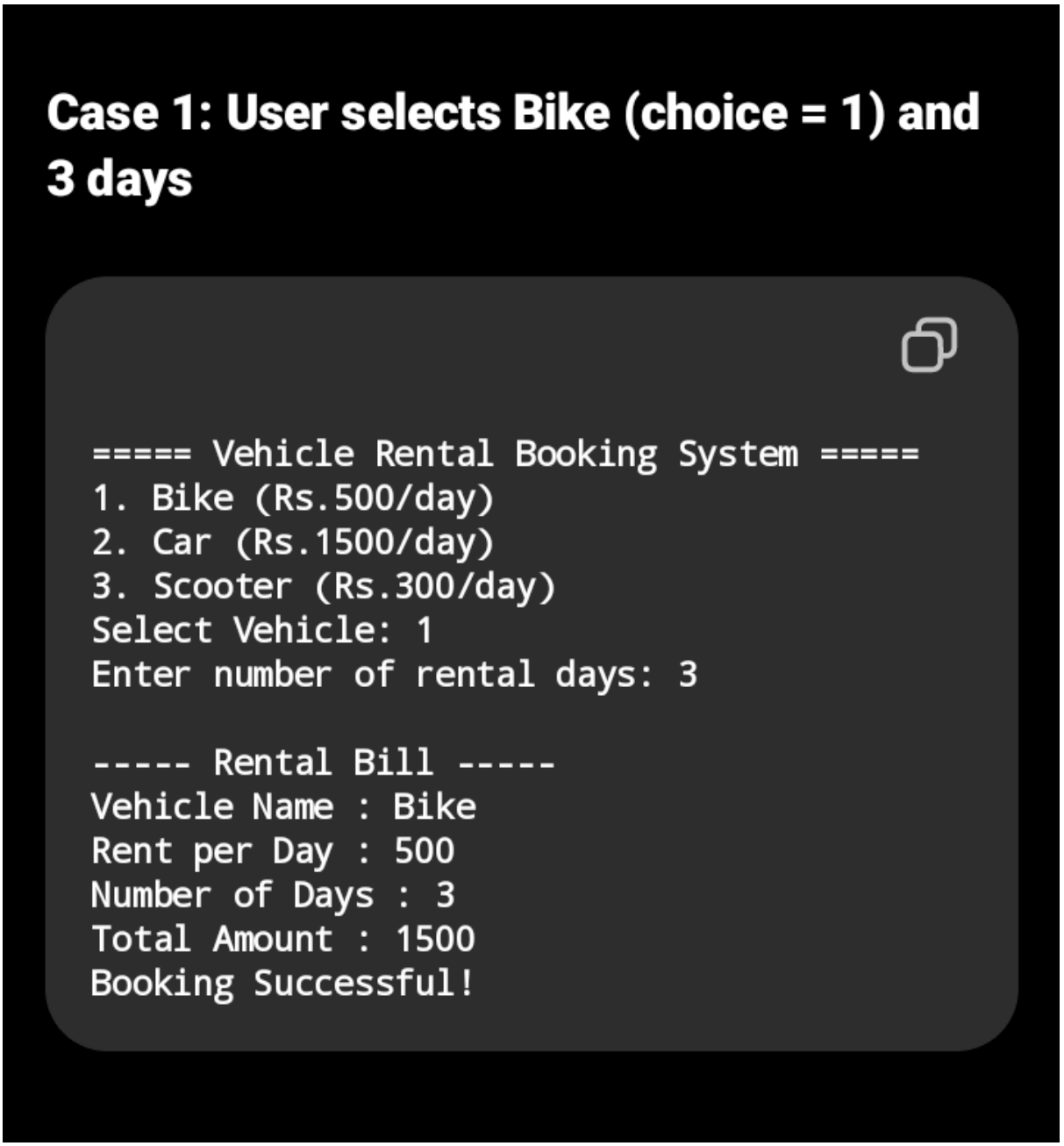
)×DailyRate.

4.7 Flowchart

5. Source Code

(In your final report, you should include the main logic for your Database Connection and the Booking Controller here.)

6. Input and Output Screens



Case 2: User selects Car (choice = 2) and 2 days

```
===== Vehicle Rental Booking System =====
1. Bike (Rs.500/day)
2. Car (Rs.1500/day)
3. Scooter (Rs.300/day)
Select Vehicle: 2
Enter number of rental days: 2

----- Rental Bill -----
Vehicle Name : Car
Rent per Day : 1500
Number of Days : 2
Total Amount : 3000
Booking Successful!
```

Case 3: Invalid choice

```
===== Vehicle Rental Booking System =====
1. Bike (Rs.500/day)
2. Car (Rs.1500/day)
3. Scooter (Rs.300/day)
Select Vehicle: 5
Invalid choice!
Enter number of rental days: 2

----- Rental Bill -----
Vehicle Name : null
Rent per Day : 0
Number of Days : 2
Total Amount : 0
Booking Successful!
```

7. Discussion and Conclusion

7.1 Discussion

The system successfully integrates payment gateways, which was the most complex technical hurdle. The use of cloud hosting allows for scalability as the vehicle fleet grows.

7.2 Conclusion

The Vehicle Rental Booking and Payment Management System provides a robust solution for modernizing the car hire industry. It ensures security, efficiency, and a better user experience.